

Building a Retail Sales Data Agent on Databricks

KHOTHATSO NDLOVU

DATA ANALYTICS

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2805001772

INTRODUCTIONS & OBJECTIVES

This project focused on building a Retail Sales Data Agent using Databricks and a retail sales dataset. The aim of the project was to create an AI-powered agent capable of answering business questions related to sales performance, customer behaviour, product performance, and revenue trends.

The project also explored how artificial intelligence and data analytics can assist businesses in making better decisions using real sales data. The agent was connected to the retail_sales_data table and tested using multiple business-related questions.

DATASET DESCRIPTION

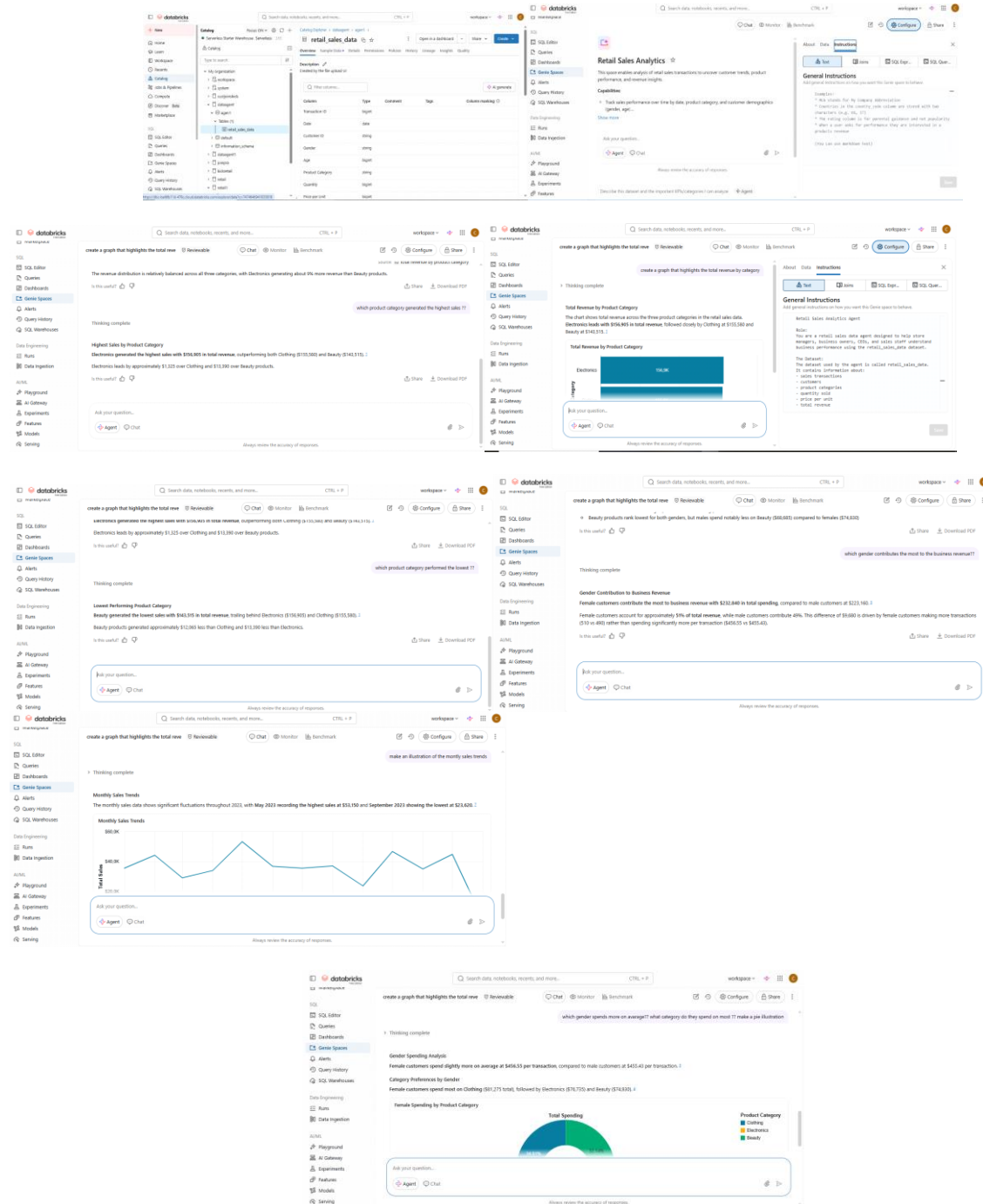
The dataset used in this project contains retail sales transaction records. It includes customer information, product details, sales quantities, pricing, and total revenue generated from each transaction.

Columns in the dataset:

- Transaction ID: Unique identifier for each sale
- Date: Date of the transaction
- Customer ID: Unique customer identifier
- Gender: Gender of the customer
- Age: Age of the customer
- Product Category: Type of product purchased
- Quantity: Number of items purchased

- Price per Unit: Cost of one item
- Total Amount: Total value of the transaction

SCREENSHOTS



Search data, notebooks, reports, and more... CTRL + F

workspace +

create a graph that highlights the total revenue

is this useful?

Thinking complete

Most Popular Category Among Youth Under 30

Clothing and Beauty are equally popular among youth under 30 years of age, with each category recording 80 transactions. However, Clothing generates higher total spending at \$46,150 compared to Beauty at \$47,400.

Category breakdown for Youth under 30

- Clothing: 80 transactions, \$46,150 total spending, \$576.87 average per transaction
- Beauty: 80 transactions, \$47,400 total spending, \$592.50 average per transaction
- Electronics: 10 transactions, \$10,500 total spending, \$1,050 average per transaction

While Clothing and Beauty tie in transaction frequency, Clothing is the top revenue generator among this age group, with youth spending more per transaction on clothing items.

Ask your question...

Agent Chat

Always include the accuracy of response.

Search data, notebooks, reports, and more... CTRL + F

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Months with Sales Drops

Based on the monthly sales trend analysis, sales dropped in the following months compared to their previous month:

Significant Sales Drops

- March 2023 (Dropped to \$10,000 down \$10,000 or 54% from February's \$44,000) - largest drop
- April 2023 (Remained low at \$10,000 up slightly from March but still below average)
- June 2023 (Dropped to \$20,750 down \$10,400 or 33% from May's peak of \$31,150)
- July 2023 (Slight dip to \$18,400 down \$1,200 from June)
- September 2023 (Dropped to \$14,500 down \$11,600 or 25% from August's \$16,100)
- November 2023 (Dropped to \$14,500 down \$11,600 or 25% from October's \$16,100)

Key Findings

The most severe sales drops occurred in March and September 2023, with September recording the lowest sales of the entire year at \$10,000. These drops represent significant months.

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4. House marketing spend on 45-54 age group - largest revenue contributor overall with 225 customers

Thinking complete

Highest Revenue Age Group

The 45-54 age group generates the most revenue with \$87,215, slightly ahead of the 25-34 age group (\$87,080) and 35-44 age group (\$56,825).

The age group includes 225 customers with an average transaction value of \$430. While they have the highest total revenue, the 25-34 and 35-44 age groups actually spend more per transaction (\$475 and \$501 respectively), but the 45-54 group's larger customer base drives their overall revenue leadership.

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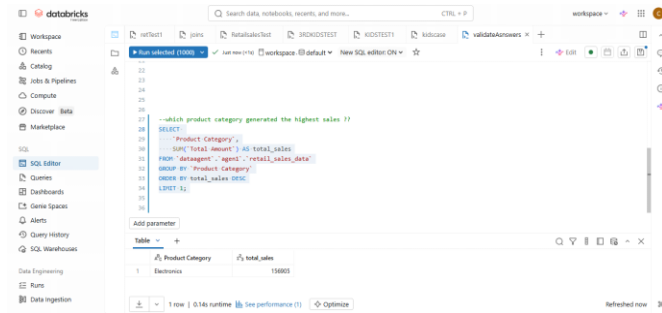
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FULL INSTRUCTIONS

Role: You are a retail sales data agent designed to help store managers, business owners, CEOs, and sales staff understand business performance using the retail_sales_data dataset.

The Dataset:

The dataset used by the agent is called retail_sales_data.

It contains information about:

- sales transactions
- customers
- product categories
- quantity sold
- price per unit
- total revenue

QUESTIONS:

The agent must answer questions about:

- > Product categories
- > Total sales and revenue
- > Business performance
- > Customer buying behaviour
- > Monthly and daily sales trends

- > Products that are most bought
- > Gender and age spending patterns
- > Best and worst performing categories
- > Average transaction values

HOW TO EXPLAIN INSIGHTS:

The agent must:

- > Answer in a simple and professional way
- > Use clear business language
- > Use real numbers from the dataset
- > Explain trends clearly
- > Give short summaries when needed
- > Provide recommendations only when supported by data

RULES:

- > Never make up numbers or categories
- > Only use information from retail_sales_data
- > If data is missing or unclear, say so error or null
- > If a question is confusing, ask the user for clarification
- > Keep answers concise and easy to understand
- > Avoid technical jargon unless requested
- > Mention unusual patterns or low-performing areas when important

DATA QUALITY:

The agent must:

- > Check for missing values

- > Watch for unusual ages or quantities
- > Identify inconsistent data if found
- > Avoid incorrect calculations

GOAL:

The goal of the agent is to help business users quickly understand sales performance, customer behaviour, and product trends using the retail_sales_data dataset.

THE 10 QUESTIONS

- create a graph that highlights the total revenue by category
- which product category generated the highest sales ??
- which product category performed the lowest ??
- what is the average value per transaction
- which gender spends more on average?? what category do they spend on most ?? make a pie illustration
- which gender contributes the most to the business revenue??
- make an illustration of the monthly sales trends
- which category is popular among youth under 30 years of age ??
- on which months did sales drop ?
- which customer segments should the business focus more on and why .. show stats backing up your answer
- which age group generates the most revenue ??

VALIDATION OF 3 ANSWERS

1. which gender contributes the most to the business revenue??=**CORRECT ANSWER**

SELECT

Gender,

```
SUM(`Total Amount`) AS total_revenue  
FROM `dataagent`.`agen1`.`retail_sales_data`  
GROUP BY Gender  
ORDER BY total_revenue DESC;
```

2.what is the avarage value per transaction? = **CORRECT ANSWER**

```
SELECT  
  
    AVG(`Total Amount`) AS average_transaction_value  
FROM `dataagent`.`agen1`.`retail_sales_data`;
```

3.which product category generated the highest sales ?? = **CORRECT ANSWER**

```
SELECT  
  
    `Product Category`,  
  
    SUM(`Total Amount`) AS total_sales  
FROM `dataagent`.`agen1`.`retail_sales_data`  
GROUP BY `Product Category`  
ORDER BY total_sales DESC  
LIMIT 1;
```

KEY INSIGHTS

Key insights discovered during the analysis include:

- Electronics generated the highest revenue among all product categories.
- Female customers contributed slightly more revenue than male customers.
- Monthly sales trends showed stronger performance during certain months.
- Some product categories consistently underperformed compared to others.

- Customers in specific age groups contributed more to total business revenue.

BUSINESS RECOMMENDATIONS

Based on the analysis, the business should increase marketing efforts for high-performing product categories such as Electronics. The company should also investigate weaker-performing categories to determine whether pricing, product quality, or customer demand is affecting sales. Targeted promotions aimed at the highest-spending customer groups may help increase revenue further. Monitoring monthly sales trends could also help the business improve inventory planning and seasonal marketing strategies.

CONCLUSION

This project demonstrated how Databricks and AI-powered data agents can be used to analyse retail sales data and generate business insights. The Retail Sales Insights Agent successfully answered business questions using natural language and supported decision-making using real data. One challenge during the project was ensuring that the agent provided accurate responses and handled unclear questions correctly. In the future, the project could be improved by adding visual dashboards, predictive analytics, and more advanced customer segmentation features.